UNIT42 · FIRMWAREX

# Firmware Security Assessment

## Sample firmware

Generated 10/20/2026, 10:26:17 AM

## Executive summary

This assessment recorded **17 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

| Critical | High | Medium | Low | Info |
|---|---|---|---|---|
| 0 | 1 | 6 | 10 | 0 |

## Scope and coverage

| Stage | Outcome | Restriction |
|---|---|---|
| extract | extracted | See scope limits below |
| static | completed | See scope limits below |
| emulation | completed | See scope limits below |
| symbolic | completed | See scope limits below |
| secrets crypto | completed | See scope limits below |
| normalize | completed | See scope limits below |

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

# Binary protection summary

| Binary | Architecture | NX | PIE | RELRO | Canary | FORTIFY |
|---|---|---|---|---|---|---|
| /bin/service-probe | x86_64 | Present | Not detected | partial | Present | Not detected |
| /bin/taint-demo | x86_64 | Present | Not detected | partial | Not detected | Not detected |

# Detailed findings

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## 1. Memory safety candidate High

| Property | Result |
|---|---|
| CWE | CWE-787 |
| Affected file | /bin/taint-demo |
| Architecture | x86_64 |
| Confidence | probable |
| Review status | new |
| Address | 0x500008 |
| Function / sink | strcpy |
| Detected | 2026-10-20T04:55:51.262761Z |

**Evidence**

| Observation | Value |
|---|---|
| condition satisfiable | true |

| Observation | Value |
| --- | --- |
| Observed taint path | 0x401136 → 0x401040 → 0x500010 → 0x40115c → 0x40116b → 0x401030 → 0x500008 |
| Reaching condition (diagnostic expression) | [{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract" … [excerpt; full expression in JSON export] |
| fault address | 0x404030 |
| overflow offset | 8 |
| stack hash | bfc322b46d82910dab6757cff82dd695801b2df2172325cb b73b4e88d0926fc2 |

Evidence reference: evidence:aaf0c444e9d234077709242be121133737482c4a2cf156b106b346e6432f090d. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

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## 2. Weak cryptography candidate Medium

| Property | Result |
| --- | --- |
| CWE | CWE-327 |
| Affected file | /etc/controlled.conf |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |

| Property | Result |
| --- | --- |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355423Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | security_weak_crypto |
| fingerprint | f5f20b3817d26c0095aa20c6ff120b80724e32967ca1a13f2e8d5e99d6f0857c |
| byte offset | 255 |
| byte length | 3 |

Evidence reference: evidence:03240ed64c0f5d0eed5614baea9aa175efea8cb419bb4d10d9ced3d3dffaf95e. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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## 3. Weak cryptography candidate Medium

| Property | Result |
| --- | --- |
| CWE | CWE-328 |
| Affected file | /etc/controlled.conf |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355402Z |

### Evidence

| Observation | Value |
|---|---|
| rule | security_weak_crypto |
| fingerprint | 93f3b3d6b9ab0007dc202fda84d078a1c25863f9cc702f61403db32c231e8597 |
| byte offset | 237 |
| byte length | 3 |

Evidence reference: evidence:ac8a8c116591d9ac08afed884b72b2d84ccf73a96da52254b64da7c9ca04ad62. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

### Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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## 4. Embedded secret candidate Medium

| Property | Result |
|---|---|
| CWE | CWE-259 |
| Affected file | /etc/controlled.conf |
| Architecture | x86_64 |
| Confidence | probable |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355267Z |

### Evidence

| Observation | Value |
|---|---|
| rule | credential |

| Observation | Value |
| --- | --- |
| fingerprint | 2dd37db128b1314e9ff9b012c0ec08b76edaeaba37c59d3a55d4f2dbf4147d03 |
| byte offset | 9 |
| byte length | 19 |

Evidence reference: evidence:bb860525034f052c47797cfcc5686f165339cf7ce2f75aa8bb1266e3d04d44d0. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 5. Embedded secret candidate Medium

| Property | Result |
| --- | --- |
| CWE | CWE-798 |
| Affected file | /etc/controlled.conf |
| Architecture | x86_64 |
| Confidence | probable |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355301Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | credential |
| fingerprint | 312863bfc75f3e34f3389259c36c57dd880f9f2147da4262f8a9b1d48b6fa6ac |

| Observation | Value |
| --- | --- |
| byte offset | 37 |
| byte length | 24 |

Evidence reference: evidence:03d514fffbcbfbe262437f7be88dfe8f524e753e7ff99023fa75279916286444. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 6. Embedded secret candidate Medium

| Property | Result |
| --- | --- |
| CWE | CWE-321 |
| Affected file | /etc/controlled.conf |
| Architecture | x86_64 |
| Confidence | probable |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355326Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | private_key |
| fingerprint | c2bde0d30690d925c8d154abb84ade64e3155b519eb24941932322fd01f05a43 |
| byte offset | 62 |
| byte length | 78 |

Evidence reference: evidence:b9625171da03865099da50023832b1c4bcef5e0f68a29c43aad9f44c66057d1e. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 7. Embedded secret candidate Medium

| Property | Result |
|---|---|
| CWE | CWE-259 |
| Affected file | /etc/redacted-377c3ecbd8de0a69 |
| Architecture | x86_64 |
| Confidence | probable |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355672Z |

**Evidence**

| Observation | Value |
|---|---|
| rule | credential |
| fingerprint | b6a307d515746ec9bbd41fd3137be82540da3b5d12ff716cdb31dfb72f7e2b1e |
| byte offset | 38 |
| byte length | 25 |

Evidence reference: evidence:1008542633036b0c9ae396b0d6f54eb8abbc6d3e1662ff6b4e16fb5620684fc2. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 8. Memory safety candidate Low

| Property | Result |
|---|---|
| CWE | CWE-676 |
| Affected file | /bin/service-probe |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | strcpy |
| Detected | 2026-10-20T04:55:46.876273Z |

**Evidence**

| Observation | Value |
|---|---|
| rule | potentially_dangerous_function_symbol |
| invocation proven | false |
| overflow proven | false |

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

## 9. Memory safety candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-676 |
| Affected file | /bin/taint-demo |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | strcpy |
| Detected | 2026-10-20T04:55:46.876474Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | potentially_dangerous_function_symbol |
| invocation proven | false |
| overflow proven | false |

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

## 10. Embedded secret candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-798 |

| Property | Result |
| --- | --- |
| Affected file | /etc/controlled.conf |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355366Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | high_entropy |
| fingerprint | 36505a624c84fd4c1ffe8fd107108d655a301d4231b4744edbfa7948278ab2d7 |
| byte offset | 29 |
| byte length | 32 |

Evidence reference: evidence:ac36c1993cd0a75b2d1b363a53b6364f98870fd4ef57cb6aaa756a29241009da. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 11. Embedded secret candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-798 |
| Affected file | /etc/controlled.conf |
| Architecture | x86_64 |

| Property | Result |
| --- | --- |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355384Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | high_entropy |
| fingerprint | b8737cd86881361823e5a2858ca8ba86b1d8ca1ba74f650d9b5ea9ccd3abd908 |
| byte offset | 259 |
| byte length | 29 |

Evidence reference: evidence:ae6d4964f34fc1cadfb188554545063a12e4948df5aae56f26e884ef6a9b0f4a. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 12. Embedded secret candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-798 |
| Affected file | /bin/service-probe |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |

| Property | Result |
| --- | --- |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.354234Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | high_entropy |
| fingerprint | 3718d99c140415f7e4510753038a82a5bc2441604c66acc9a5b5b8cfec8b8280 |
| byte offset | 511650 |
| byte length | 50 |

Evidence reference: evidence:1ebc6dd22b2c55b05fd4a8a6c4ba1ded96d9cd9d2d84b671fa489d4ccdce1962. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 13. Embedded secret candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-798 |
| Affected file | /bin/service-probe |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |

| Property | Result |
| --- | --- |
| Detected | 2026-10-20T04:55:51.354304Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | high_entropy |
| fingerprint | 5246c733aaababa44755c35a5e66dc68128c5e57a62f0a17e6b52ec9812b7cd5 |
| byte offset | 591137 |
| byte length | 26 |

Evidence reference: evidence:a8b03cb4a30a7765f189f879067b8636b4b10d72d57671a848996898f92f9e66. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 14. Embedded secret candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-798 |
| Affected file | /bin/service-probe |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.354329Z |

### Evidence

| Observation | Value |
| --- | --- |
| rule | high_entropy |
| fingerprint | f616b2cda1343dbceff1c0b5526326c8fed9c2e62dce2359fe511f983e2f2a02 |
| byte offset | 591169 |
| byte length | 26 |

Evidence reference: evidence:13fcd4309a8b21b528eefe56baeaef5d766879052bd4ca9755f4644ddaa8b810. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

### Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 15. Embedded secret candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-798 |
| Affected file | /bin/service-probe |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.354358Z |

### Evidence

| Observation | Value |
| --- | --- |
| rule | high_entropy |

| Observation | Value |
| --- | --- |
| fingerprint | 20a69671db6fec6c47315450b754b3bc4f0a53f0c3dca2f360135722261dcac3 |
| byte offset | 591328 |
| byte length | 36 |

Evidence reference: evidence:bee33a81f7dc2b2b02492a72774770e403f7780df5f344ef8cdb194e7e0887b5. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 16. Embedded secret candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-798 |
| Affected file | /bin/service-probe |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.354378Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | high_entropy |
| fingerprint | bf9cc91c167091c0137c539f9636cae1f06158d788804e71d0568fbb571ff29d |

| Observation | Value |
| --- | --- |
| byte offset | 591392 |
| byte length | 36 |

Evidence reference: evidence:7d54ba10f1e165bc5229f32c0a521e89942abde96e739497790b7271c7db8549. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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## 17. Embedded secret candidate Low

| Property | Result |
| --- | --- |
| CWE | CWE-321 |
| Affected file | /etc/controlled.conf |
| Architecture | x86_64 |
| Confidence | heuristic |
| Review status | new |
| Address | Not established |
| Function / sink | Not applicable |
| Detected | 2026-10-20T04:55:51.355345Z |

**Evidence**

| Observation | Value |
| --- | --- |
| rule | certificate |
| fingerprint | 25df859d37c98ed033a6b83ed06b583fc94ff5d7eb5bdae876205d1006edce1c |
| byte offset | 141 |
| byte length | 75 |

Evidence reference: evidence:2c0da87793a65f2e3318022d7dd799f396b5b993b68edc12cacdc5588520396a. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

**Recommended next step**

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

# Assessment identity

| Property | Value |
| --- | --- |
| Firmware ID | 6f336cad-c57c-4d22-9872-69ae6aed41f0 |
| SHA-256 | a71d1e4937d2eb468ea965462c27f6b3c99eb2207fa39bd59d04161bf45856c9 |
| Job ID | 923f5f44-7704-472a-9f1f-5c3e227ffe41 |
| Image bytes | 339968 |